NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

MS0003

# Introduction to Data Science and Artificial Intelligence

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Statistical Description:

Univariate Statistics

- Mean
- Standard Deviation
- Median
- Quartiles

Analytical Visualization:

- Box Plots
- Histograms and KDE
- Violin Plots

# Box Plot

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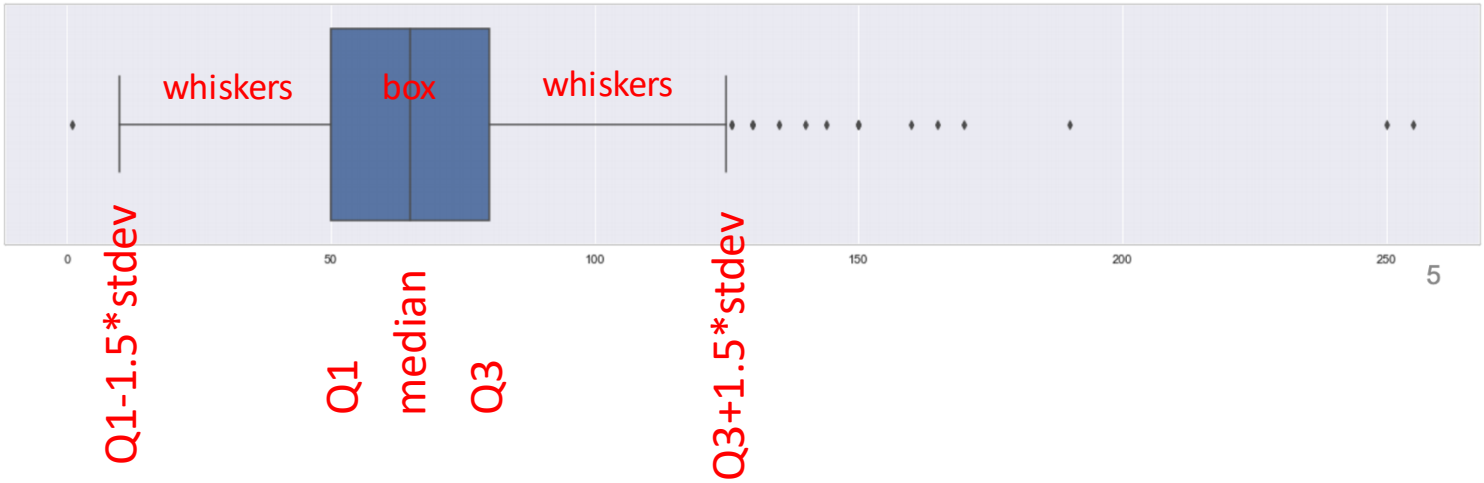
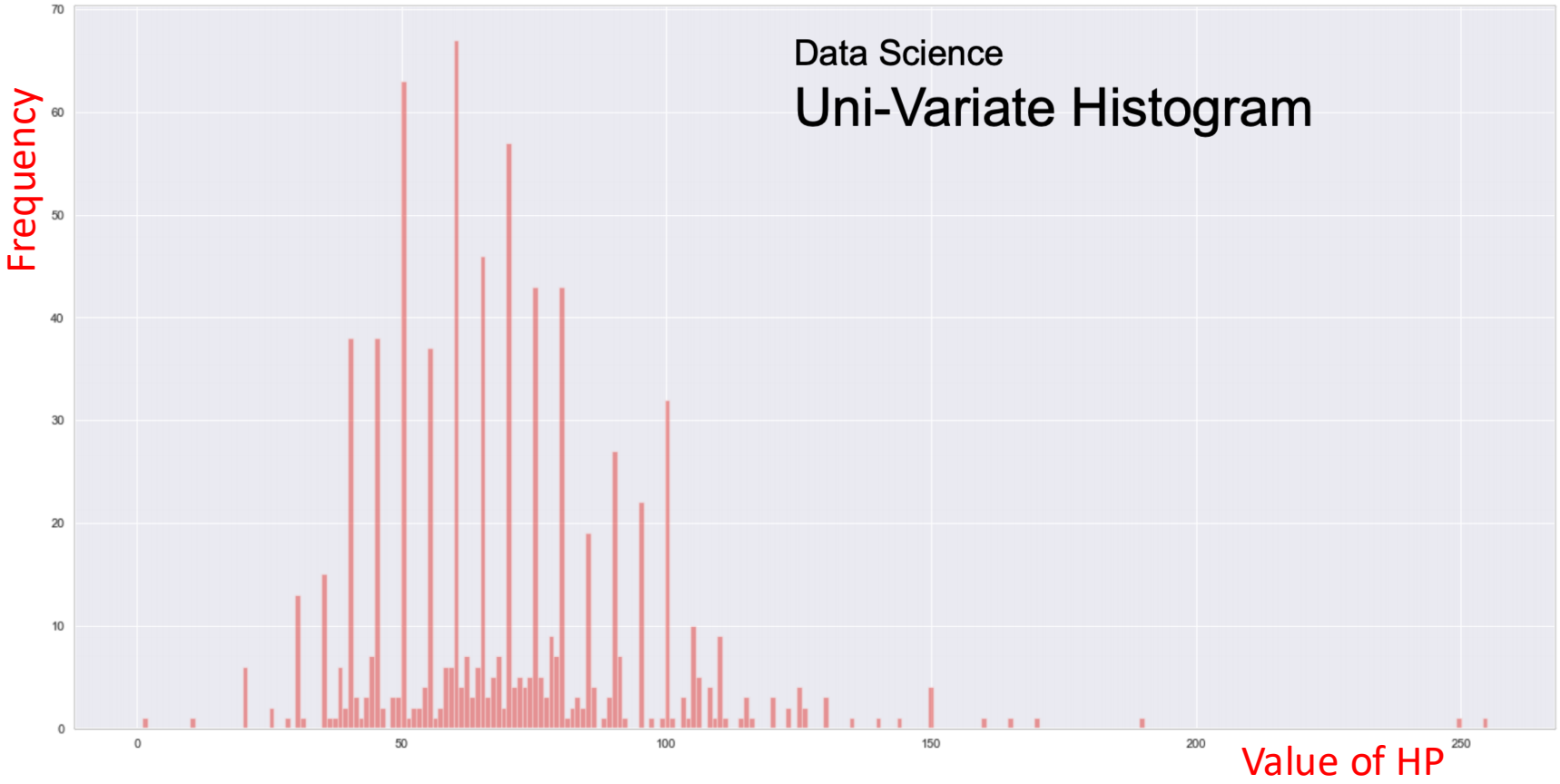
## Uni-Variate Box-Plot

| | |
|---|---|
| **min** | 1.000000 |
| **25%** | 50.000000 |
| **50%** | 65.000000 |
| **75%** | 80.000000 |
| **max** | 255.000000 |



# Histogram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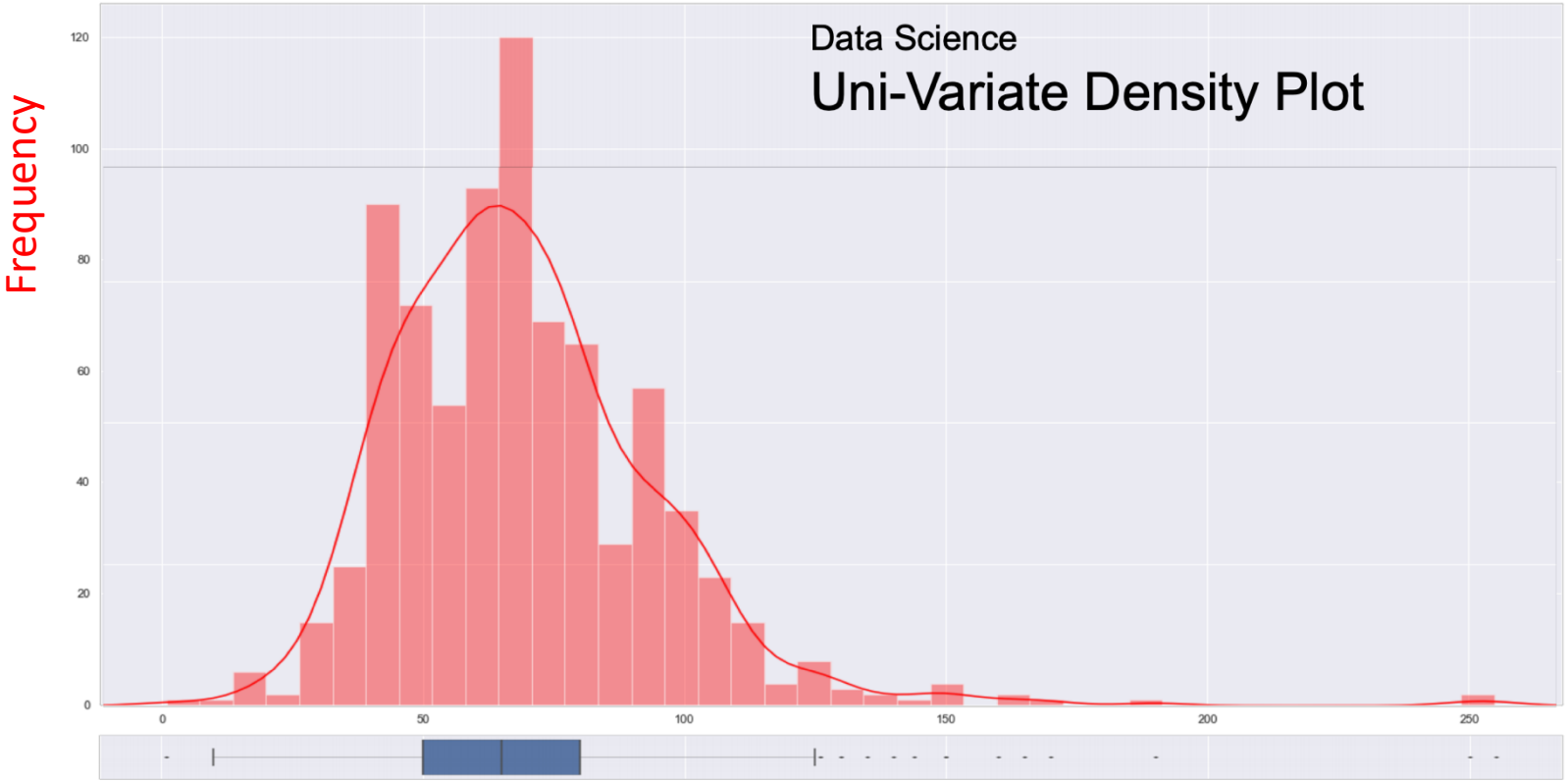
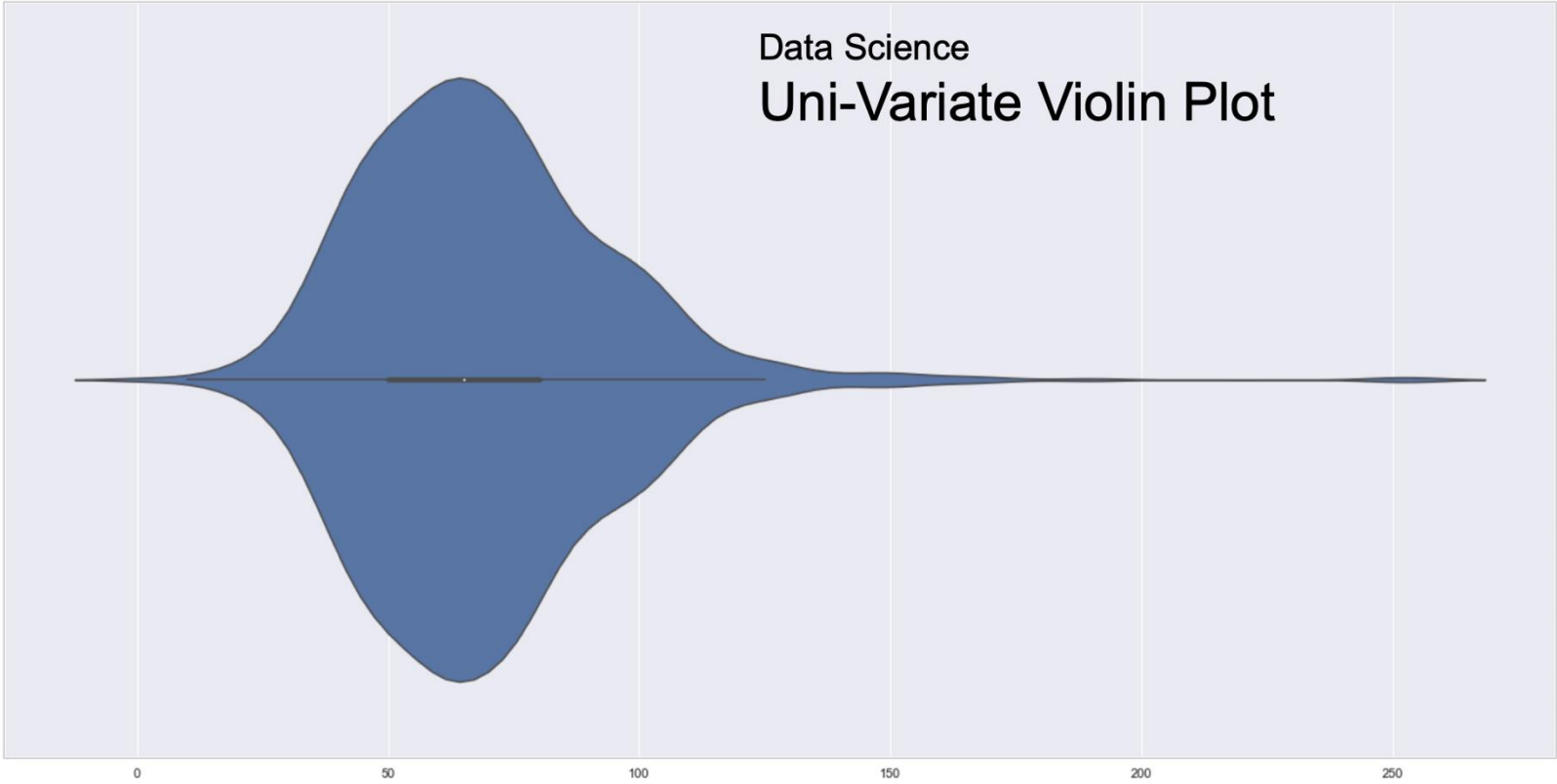
Uni-Variate Histogram
Frequency
70
60
50
40
30
20
10
0
0
50
100
150
200
250
Value of HP

# Binning and Density Plot - kernel density estimation (KDE)



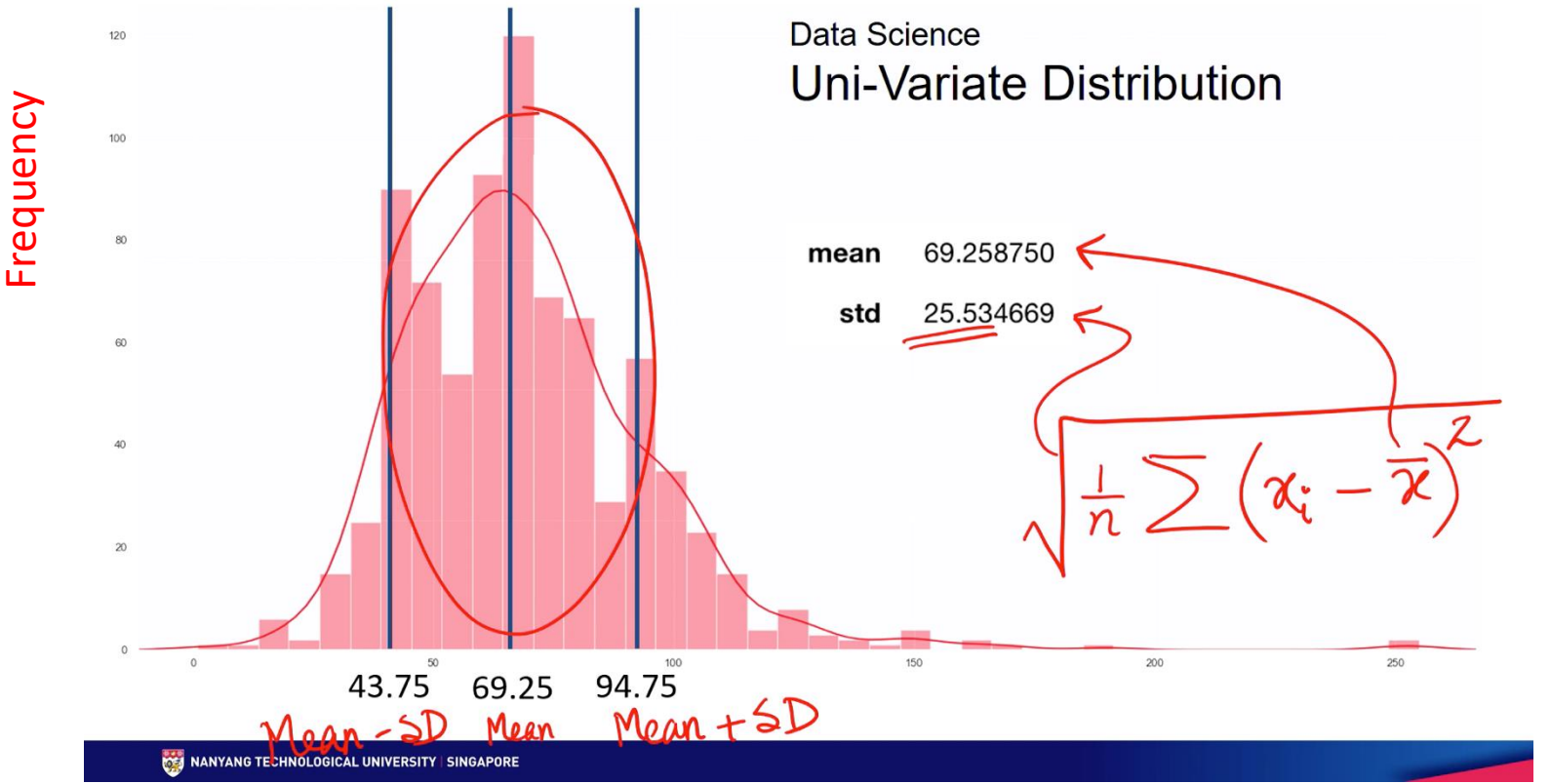
# Violin Plot = *Box* + *Density*

Frequency

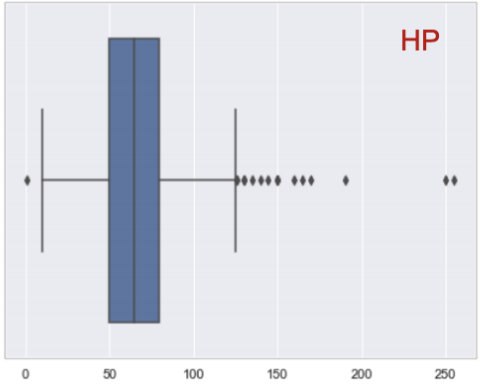


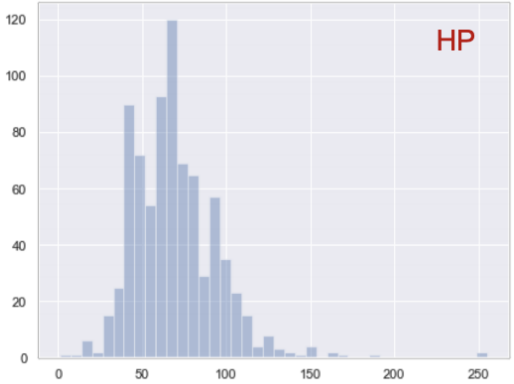

Value of HP

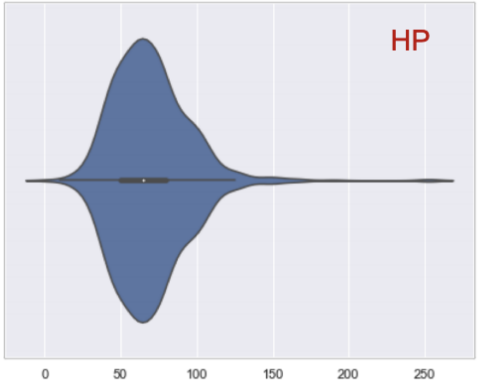
# Mean and Standard Deviation

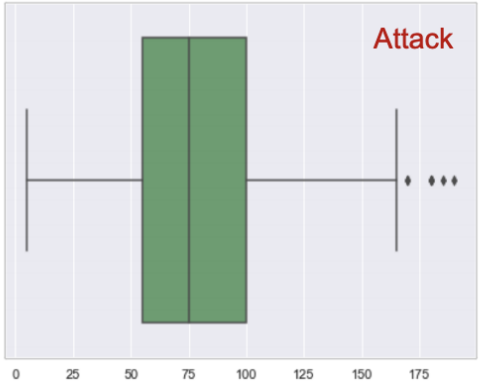


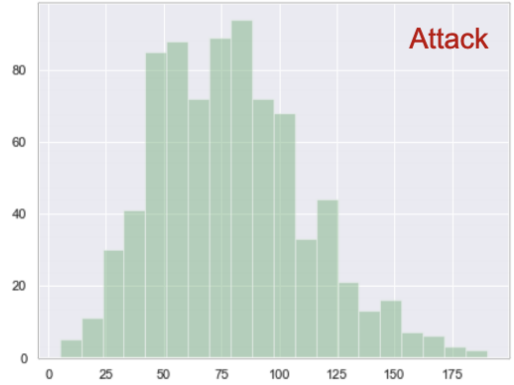
# Bivariate Statistics Visualization (*x2!*)

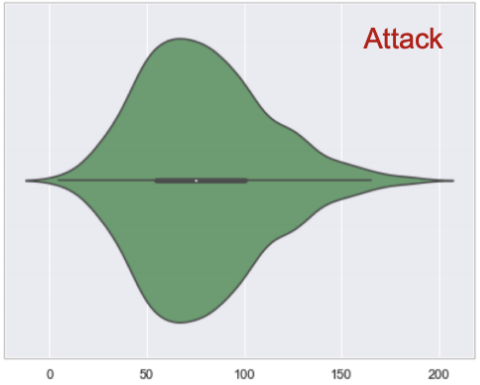


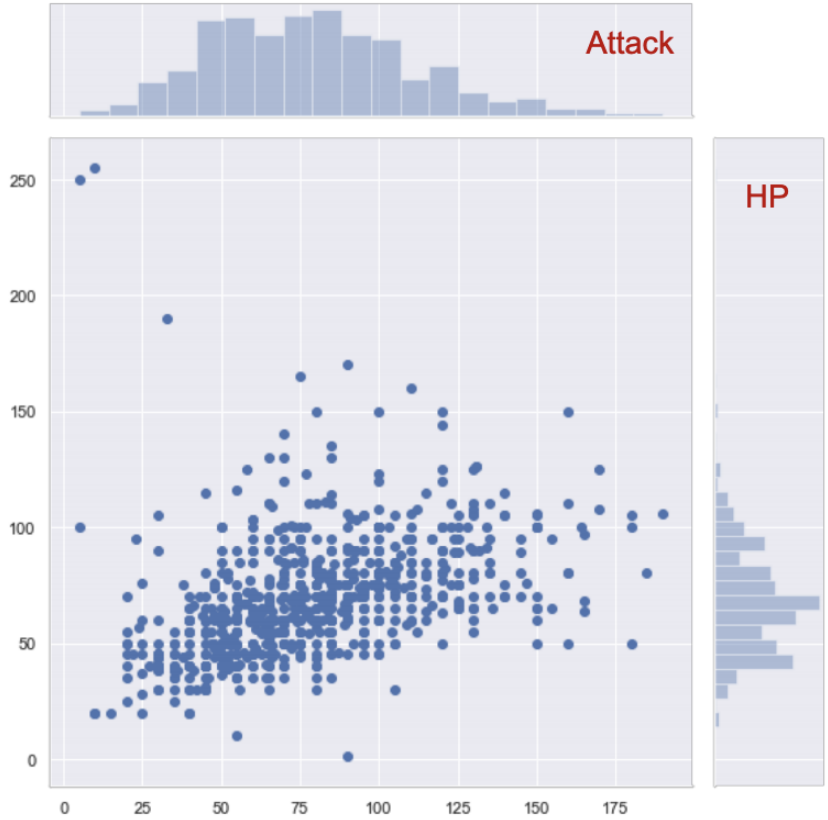



## Pattern Recognition:

Bivariate Statistics

- Statistical Relation between two variables

# Joint Plot or a Pair Plot

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## Bi-Variate Joint Plot

**Statistical Relation**

| | |
|---|---|
| **HP** | Plotted along Y axis |
| **Attack** | Plotted along X axis |

**Pearson's Correlation Coefficient (between -1 to +1)**

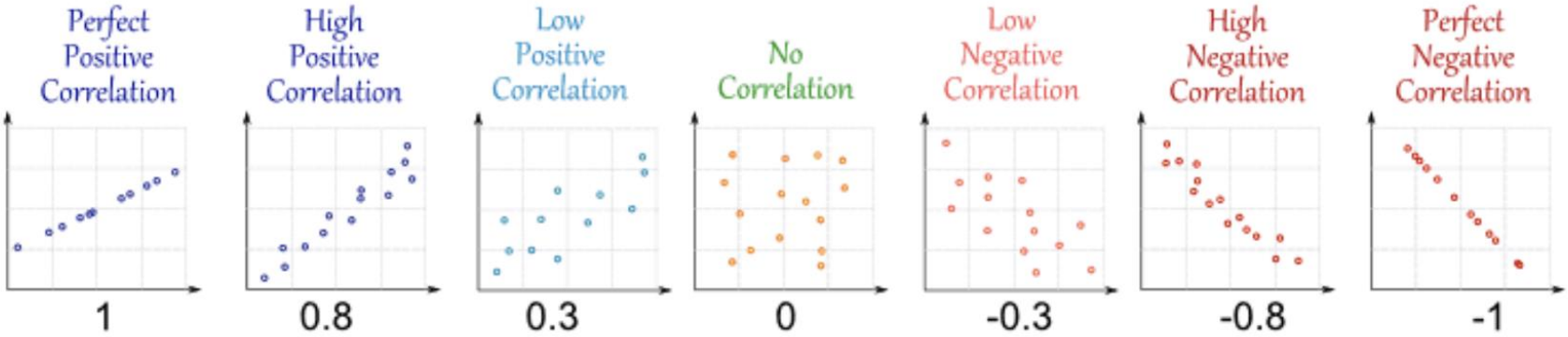
**Co-Variance / St. Dev Product**

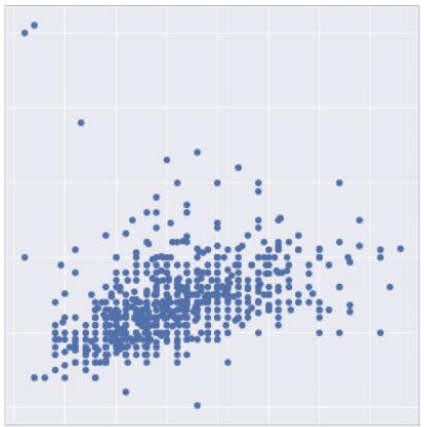
$$\rho_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}}$$

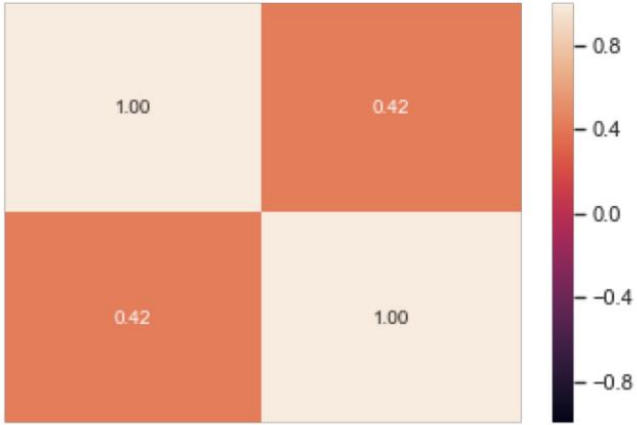
Cov

StDev

# Correlation Matrix (heat map)






Play this to improve your Correlation spotting skills

**Guess the Correlation**

http://guessthecorrelation.com/

# Multi-Variate Correlation Matrix (heat map)

**Mutual Correlations**

No Dependence

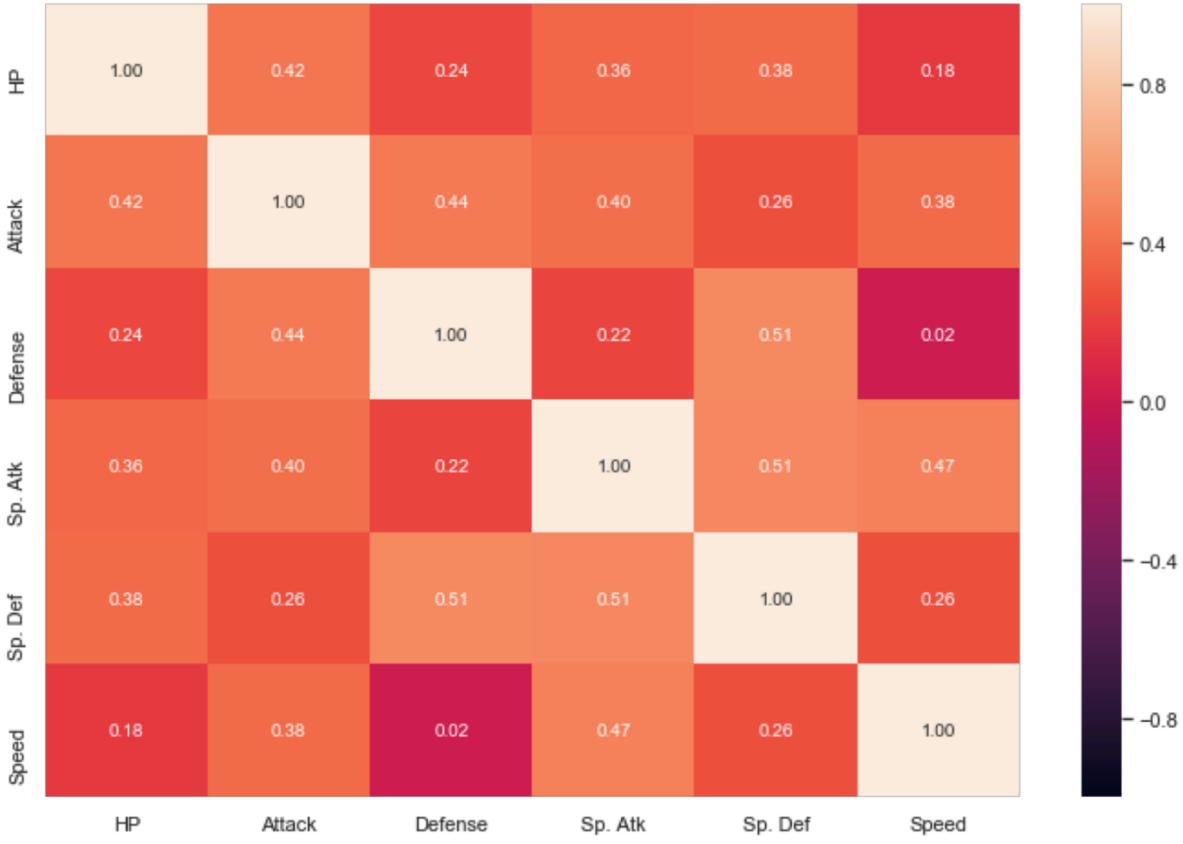
Corr = 0

Perfect Positive

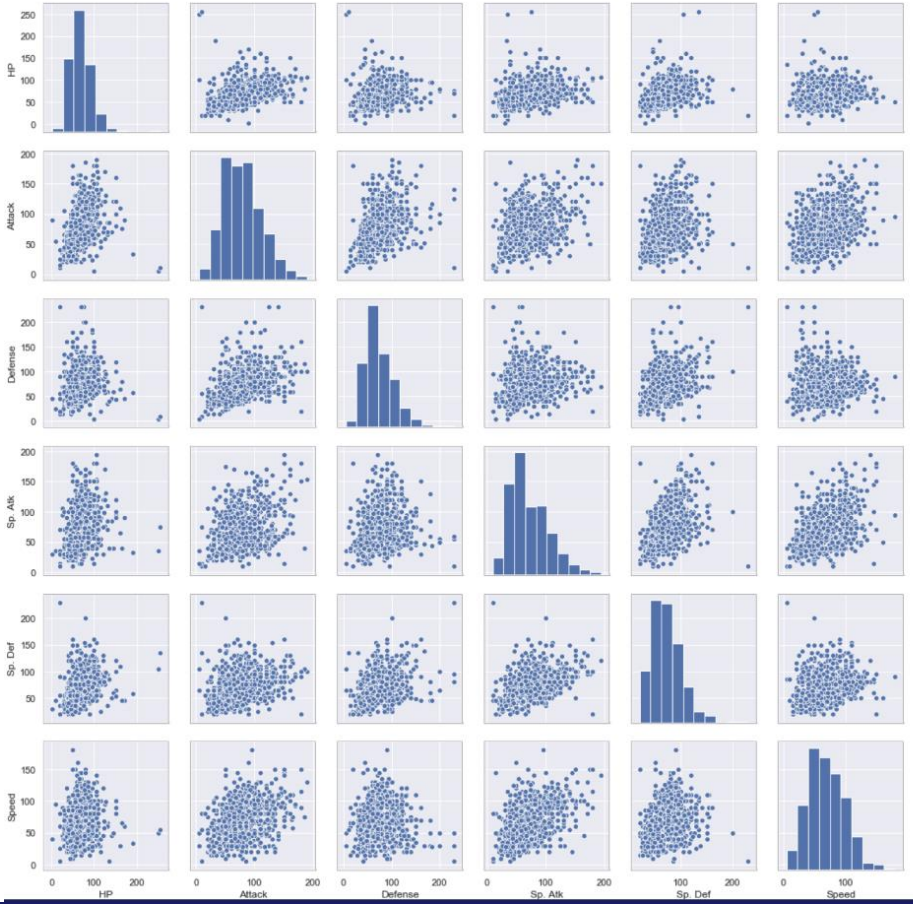
Corr = + 1

Perfect Negative

Corr = - 1



# Multi-variate Pair Plot

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## Multi-Variate Dist.

**Pair-Plot of Multi-Variate Data**

Histograms in the Diagonals

Scatter-Plot in Off-Diagonals

Pattern recognition

- Distributions of Variables
- Inter-Variable Dependence